Draw only the forces on your chosen object

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Free-body diagram / ES: diagrama de cuerpo libre / PT: diagrama de corpo livre

Only the forces acting on one chosen object.

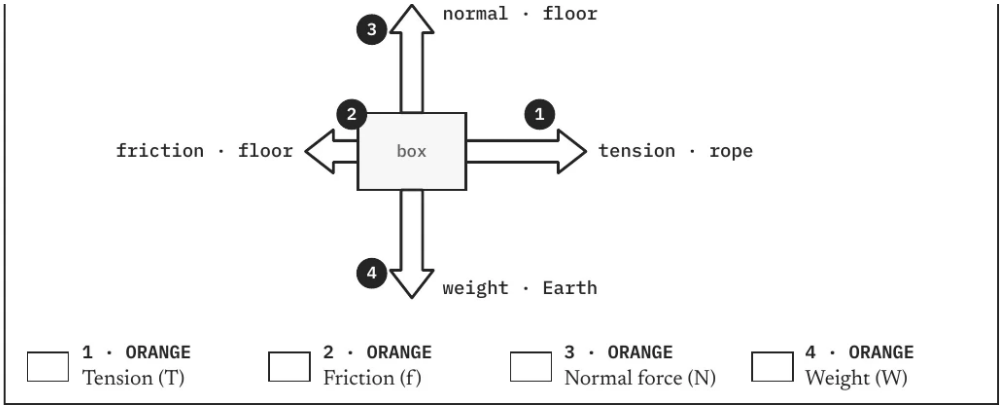
Also on your note page: Selected object, Interacting object, Force source.

B READ THE RELATIONSHIP

A free-body diagram shows the external forces on one chosen object.

Choose the receiver before drawing forces

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces.

Given weight = 20 N at rest one rope

Rule sources: Earth, rope

Work weight 20 N down tension up, balances it

Answer Tension 20 N up equal-length arrows

E SAVE FOR THE FINAL MEETING

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

Read the physics: Free-body diagram

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Draw a dot or box for the object. Name each outside object that pushes or pulls it. Draw one arrow for each force on your chosen object.

02 HOW IT WORKS

Label each arrow with the source or force name. Longer arrows mean stronger forces when the scale is the same. Put velocity outside the force diagram.

03 WHAT TO CHECK

A hanging lamp has gravity from Earth and tension from the rope. Equal forces balance on the lamp. They are not a third-law pair: both act on the lamp.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces. Tension 20 N up equal-length arrows.

05 CONNECT TO XJ

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

SAY IT BEFORE YOU WRITE

Which object receives every force in this diagram? I drew ___ because ___ acts on the chosen object.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Free-body diagram / ES: diagrama de cuerpo libre / PT: diagrama de corpo livre

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning To me, free-body diagram means …	**How can you tell?** If the lamp is your object, which arrows belong on its diagram? I include the forces that act on ___.
Example or sketch I see it when …	**Non-example** This looks close, but …

B DRAW WITH YOUR TEACHER

Draw a boundary around one object. Add force arrows with source labels; put motion information outside the diagram.

Labels you can use: object · boundary · outside.

Chosen object: _____

Tell a partner: I drew ___ because ___ acts on the chosen object.

On assignment: Free-body diagram

SECTION E / BACK / FINAL MEETING ONLY Name _____

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Systems engineer**

Choose the boundary of the simulated object. Draw only the force layers you would send to the acceleration calculation.

☐ **Animation director**

Make a two-layer storyboard: physical forces on XJ, and motion. Tell the animator which Earth arrow belongs in another shot.

Word bank: object · boundary · outside.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”